

The reconstruction of the ASEAN Digital Integration Index

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1 The ASEAN Digital Integration Index

The ADII was created by the ASEAN Economic Community in 2020-2021 as part of the efforts to strengthen and integrate the digital development of the region in the face of the COVID-19 pandemic. In 2022-2023, it was updated to version 2.0 to provide a comprehensive update on the digital transformation of the ASEAN Member States. The index consists of six pillars, representing different domains of digital development, but focusing strongly on the role of technology in the economy.

The indicators making up the index were selected based on six key criteria:

- Relevance - Data must be clear, accurate, and directly aligned with Digital Integration Framework priority areas.
- Accessibility - Data must be easily obtainable and made available without cost.
- Coverage - Indicators must encompass at least eight of the ten ASEAN Member States.
- Timeliness - Datasets must be recent, typically published from 2017 onwards.
- Consistency - Data sources must be published on a regular and predictable basis.
- Data must originate from reputable sources with open methodologies and disclosed limitations.

Although the ADII was created with ASEAN states in mind, these selection criteria make it suitable for adoption in other parts of the world. Most of the data is sourced from global, publicly available sources, making it easier to extend outside of the region. This is in contrast to the DESI index, in which some indicators are collected solely for this purpose by the EU Member States institutions. On the other hand, although selection criteria

included Timeliness and Consistency, institutions providing the data may discontinue their selection. For example, several indicators come from the World Economic Forum's Global Competitiveness Report, which was not updated since 2019.

The six pillars of the ADII map precisely to the priority areas of the ASEAN Digital Integration Framework (DIF), which regional leaders have identified as essential for a unified digital economy. Each pillar consists of 5 indicators, although some of them may be aggregations of several simpler measures. The indicators are transformed to a scale between 1 to 20, and each pillar is the sum of the maximum value of 100. The final index is the average of all its pillars.

Pillar 1: Digitally Enabled Trade & Logistics - Assess the adoption of digital technologies and infrastructure used to facilitate seamless cross-border trade. The digitalisation of this sector may be less important for economies less reliant on international trade. However, part of the indicators can relate to transport and trade of goods within the country.

Pillar 2: Data Protection & Cybersecurity - Measure the degree to which frameworks protect data privacy while encouraging secure digital innovation. This is an important safeguard ensuring the fair adoption of digital technologies and a big benefit over other popular indicators (such as European DESI) that do not include cybersecurity indicators.

Pillar 3: Digital Payments & Identities - Assess the adoption and interoperability of digital financial services and national identity systems. Digital payments have been quickly growing in popularity in the region, as e-commerce is the main area of ASEAN's digital economy (Rumata and Sastrosubroto, 2020).

Pillar 4: Digital Skills & Talent - Quantify the availability of a skilled workforce capable of driving the regional digital economy. The possibility of actually making use of digital advancements and quickly switching to new technologies in the case of a crisis is dependent on human skills and attitudes. Moreover, education and knowledge is a crucial area due to its contribution into the digital divide (Lythreath et al., 2022).

Pillar 5: Innovation & Entrepreneurship - Examine the business environment's ability to foster and scale digital start-ups and IP creation, the availability of financing and ease of starting a business.

Pillar 6: Institutional & Infrastructural Readiness - Determine the public sector's responsiveness to digital change and foundational infrastructure availability. This pillar includes both hardware infrastructure and laws and institutions.

The indicators included in the ADII come from seven sources, listed in Table 1.

The **OECD Trade Facilitation Indicators** is a database maintained by the Organisation of Economic Cooperation and Development. It contains information on movement, release, and clearance of goods at the border in 163 countries. The **World Bank Open Data** is the source maintained by the World Bank Group, providing a wide selection of global development data. From this source, ADII uses the Logistic Performance Index, Global Financial Development database, and Identification for Development database. The World Bank also provides access to Global Competitiveness indicators by the World Economic Forum. Another global source is the ITU Global Cybersecurity Index, created by the **International Telecommunication Union**. These data are supported by the Data Protection Index from the 2020 report prepared by the **TRPC**. It was, however, only collected for 30 countries, mainly from the ASEAN. ADII also sources indicators from the **UNESCO Institute for Statistics** database, the Global Innovation Index report by **WIPO**, and the **United Nation's** Global Survey on Digital and Sustainable Trade Facilitation.

The ADII was released in two editions, the first in 2021, and the second in 2023. In the first edition, pillars 2 (cybersecurity) and 6 (institutional and infrastructural readiness) received the highest scores. But in the second edition, ASEAN countries rapidly improved scores in the first pillar (related to digital payments and identities), probably due to e-commerce's increasing popularity after the pandemic. Pillar 4 (digital skills) has been and remains the least developed area, while Pillar 5 (innovation and entrepreneurship) is only slightly better.

2 Data collection

First, I identified the indicators within the sources and collected the data. The sources were either in a form of database, where downloading the data was straightforward, or a report, where I scraped the data using NotebookLM tools. Table 1 details sources and issues encountered during the collection exercise. In a few cases, the source contained several indicators that could match the description in ADII report, and it was not clear which should be included.

Indicator 1.1 is a compound index describing how trade and customs processes are supported by digital technologies. Among the OECD Trade Facilitation Indicators, I identified several indicators that matched this description and selected five with enough data coverage: G4 (electronic pre-arrival processing), G6 (Processing system electronic payment), G9 (electronic data interchange), G10 (automated processing - goods release) and G12 (full-time

automated processing). For the Indicator 1.3 I chose variable A7 from the OECD Trade Facilitation Indicators database, which describes Import/Export procedures.

According to the ADII report, indicator 3.4 should come from the World Bank's Global Financial Development (Findex) database, but I found this variable in the World Bank's Identification for Development database. Given that this is the same institution, it could have been a simple mistake in the report. Another indicator from this database (3.5) is supposed to describe the degree to which a digitized ID system is in place. I found two variables regarding digitised ID, one for the percentage of the population using it, and the second being a binary variable with 1 if the country has a digitised ID and 0 otherwise. I chose the latter because of much better data availability, although none of them seems to directly match the description.

In three cases, the original dataset was very limited, and I used other sources to expand it. The indicator 5.5 (protection of intellectual property) was available for only 53 countries. When missing, I supplemented it with the same variable from the World Bank database [detailed name]. When indicator 3.2 (using digital technologies for financial transactions) was not available, I copied the value for the country from the indicator 3.1 (using digital technologies for banking), because digital banking could be categorised as part of the wider financial uses. However, it can mean that indicator 3.2 is underestimated in the dataset.

The indicator 6.1 (active mobile-broadband subscription) was taken from the same source, but from the ITU Data Hub database instead of the report. For indicator 6.2 (proportion of internet users), I found the same variable from the World Bank database.

The most difficult case was with the indicator 2.1 (data protection), originally sourced from the TRPC report which only included 30 countries. I reconstructed this variable by using the DLA PIPER database of legal data protection measures and the questionnaire included in the original report. This could introduce some bias, as the the protection measures may have changed through the years. [check how similar it is for the countries I do have]

Table 1: Sources of data and issues encountered during the collection process.

Indicator	Description	Source	Issues
1.1	Degree to which trade/customs processes are supported by digital technologies	OECD Trade Facilitation Indicators	Not sure which measures are aggregated into this indicator
1.2	Degree to which digital certificates and signatures are in place	OECD Trade Facilitation Indicators	
1.3	Degree to which international standards for trade documents and procedures are followed	OECD Trade Facilitation Indicators	
1.4	Level of quality of trade- and transport-related infrastructure	World Bank Logistic Performance Index	
1.5	Level of competence and quality of logistics services	World Bank Logistic Performance Index	
2.1	Degree to which data protection measures are in place	TRPC Data Protection Index	TRPC only provided data for 30 countries, reproduced the rest
2.2	Level of legislative and regulatory cybersecurity capabilities	ITU Global Cybersecurity Index	
2.3	Level of institutional cybersecurity capabilities	ITU Global Cybersecurity Index	

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Indicator	Description	Source	Issues
2.4	Level of technical cybersecurity capabilities	ITU Global Cybersecurity Index	
2.5	Level of international cooperation on cybersecurity	ITU Global Cybersecurity Index	
3.1	Proportion of people who use digital platforms or devices for banking purposes only	World Bank, Global Financial Development (Findex) database	
3.2	Proportion of people who use digital platforms or devices for any type of financial transaction	World Bank, Global Financial Development (Findex) database	If not available, used 3.1
3.3	Degree to which legal frameworks enable electronic transactions	UN Global Survey on Digital and Sustainable Trade Facilitation	Got the same indicator from OECD Trade Facilitation Indicators
3.4	Proportion of people who have a national identity card	World Bank, Global Financial Development (Findex) database	Found this in: World Bank, Identification for Development
3.5	Degree to which a digitized ID system is in place	World Bank, Identification for Development	Only found a binary variable
4.1	Proportion of graduates in science, technology, engineering, and mathematics	UNESCO UIS database	

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Indicator	Description	Source	Issues
4.2	Proportion of employment in knowledge-intensive services	WIPO, Global Innovation Index (GII) database 2023	
4.3	Level of multi-stakeholder collaboration in R&D	World Economic Forum WEF), Global Competitiveness Report	
4.4	Degree to which the active population has digital skills	World Economic Forum WEF), Global Competitiveness Report	
4.5	Degree to which graduates have business-relevant skillsets	World Economic Forum WEF), Global Competitiveness Report	
5.1	Degree to which venture capital is available	World Economic Forum WEF), Global Competitiveness Report	
5.2	Proportion of GDP expenditure on R&D	WIPO, Global Innovation Index (GII) database 2023	

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Indicator	Description	Source	Issues
5.3	Degree to which innovative companies can grow	World Economic Forum WEF), Global Competitiveness Report	
5.4	Degree to which it is easy to start a business	World Bank, Ease of Doing Business Index	
5.5	Degree to which intellectual property protection frameworks are in place and are enforced	Global Innovation Policy Center (GIPC), International IP Index (IIPi)	53 countries only, but substituted with the same variable from the World Bank
6.1	Proportion of Active mobile-broadband subscriptions	ITU, Measuring the Information Society Report 2018	Got data from ITU Data Hub from 2018 or before
6.2	Proportion of Internet users	ITU, Country ICT Database	Did not find the data, used the same variable from the World Bank
6.3	Degree to which government services are available and accessible digitally	UN e-Government Knowledge base	
6.4	Degree to which a government is considered responsive to disruption and change	World Economic Forum WEF), Global Competitiveness Report	

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Indicator	Description	Source	Issues
6.5	Degree to which a legal framework is considered conducive for digital innovation	World Economic Forum WEF), Global Competitiveness Report	

2.1 Constructing the scores

Following the procedure described in the ADII report, each variable was rescaled to take values between 0 and 20 by dividing it by the 'top value' and multiplying by 20. In most cases, the indicators were initially created in a pre-defined scale, and the 'top value' was the maximum. If the variable was in percentages, the top value was set to 100. Indicator 6.1 (mobile-broadband subscription per 100 people) did not have a maximum value because one citizen can have more than one subscription, and the value exceeded 100 in many developed countries. I could set the 'top value' at the maximum available in the dataset (as it was done in the ADII report), but this solution had some drawbacks. First, the distribution was highly skewed and the resulting indicator would have a majority of the countries with very low scores, even if almost all citizens had subscriptions. Second, it would not reproduce the scores in the ADII report, because the maximum value among just the ASEAN sample would be different than in the whole dataset. Finally, I capped the values at 100 and set that as the top value for rescaling.

Each pillar is constructed as a sum of five underlying indicators, so the pillar scores fall naturally on a scale of 0-100. The final index is computed as an average from six pillars, preserving that scale. In the ADII report, if an indicator was missing, the pillar was computed by summing other indicators and rescaling. Since I had many countries with too many missing variables, I chose only those with at most three indicators missing. This way more countries could be included with the integrity of scores preserved as much as possible.

3 Comparison with the original scores

Finally, I was able to construct the ADII scores for 108 countries. Out of 16 countries from the original reports, 14 are included in my dataset, and only the data for Brunei Derussalam and Myanmar was missing. I compared the pillar scores and found an average of 12.3% error, with the biggest disparities for smaller economies (e.g. 33.6% for Laos and 22.3%

for Cambodia). More details of the comparison can be found in Table 2. The correlations between original and reconstructed pillar scores are high, with the lowest of 0.78 for pillar 4 (digital skills), and the correlation for the aggregated index reaching 0.98. The average errors (where errors are not treated as absolute distance between original and reconstructed value, but as simple difference) show that the reconstructed scores are usually smaller than the ones obtained by ASEAN. The only exception was the first pillar, with a slightly negative average error.

The analysis of mean percentage error shows that the biggest discrepancy is in the Pillar 3 (digital payments and identities), where the reconstructed scores underestimated values reported in the ADII report. This pillar includes several indicators that I had trouble to match (3.3 and 3.5), and indicator 3.2 with a very limited data coverage. This is likely what is causing the issues.

Pillar	Correlation	Average pct. error	Average error
P1	0.96	10.14	-0.63
P2	0.97	7.01	1.14
P3	0.86	22.62	9.52
P4	0.78	12.35	4.38
P5	0.86	13.04	5.36
P6	0.93	8.86	5.55
Index	0.98	7.56	4.22

Table 2: Comparison of recreated pillar scores with original values from the ADII report (ASEAN, 2023) $N = 14$.

4 Results

4.1 Digitalisation across the world

The full table of country scores can be found in the Appendix. The highest scores belonged to Singapore (86.5), Finland (85.9) and Denmark (83.6), followed closely by the United States (83.5), showing the the digital frontiers could developed on many continents. However, Figure 1 and Table 3 show that in regional analysis, Europe had the highest average and the most equal scores, while the Americas and Asia are very diverse in the level of development. African countries scored the lowest across all pillars, but also with mostly small standard deviations, with the exception of Pillar 2. Note that due to the fact that the dataset contains only two countries from Oceania (Australia and New Zealand), they were included in Asia, which could have slightly overestimate the regional mean.

Region	All	Africa	Americas	Asia and Oceania	Europe
N	108	24	18	33	33
Pillar 1	66.0 (20.7)	41.70 (11.10)	67.70 (16.60)	68.1 (18.8)	80.7 (13.4)
Pillar 2	82.9 (19.5)	74.00 (23.10)	72.80 (20.00)	85.2 (18.9)	92.7 (9.91)
Pillar 3	66.0 (27.6)	36.40 (19.50)	65.10 (22.30)	66.5 (24.0)	87.5 (17.4)
Pillar 4	41.7 (9.73)	33.70 (6.23)	37.1 (8.19)	44.1 (8.94)	47.6 (8.45)
Pillar 5	46.7 (14.6)	37.50 (8.40)	38.4 (13.3)	49.6 (14.1)	55.0 (13.7)
Pillar 6	59.0 (16.9)	39.00 (10.80)	56.2 (13.0)	63.7 (14.8)	70.5 (9.97)
Index	60.4 (15.8)	43.70 (10.50)	56.2 (13.1)	62.9 (13.7)	72.3 (10.4)

Table 3: Average scores per region, standard deviation in the brackets.

Figure 1 indicates as well that the countries with missing data concentrate in Africa, Middle East, Central Asia and Eastern Europe. Therefore, the regional comparisons should be interpreted with caution. Lack of data from developing nations is a big problem in global digitalisation studies, as this is both where the transition could make a big change to the economy, and where the digital divide is likely to be deeper (Mubarak et al., 2020). However, these countries often do not yet have the institutions or culture of collecting and openly sharing the data (Oloyede et al., 2023).

Disclosures

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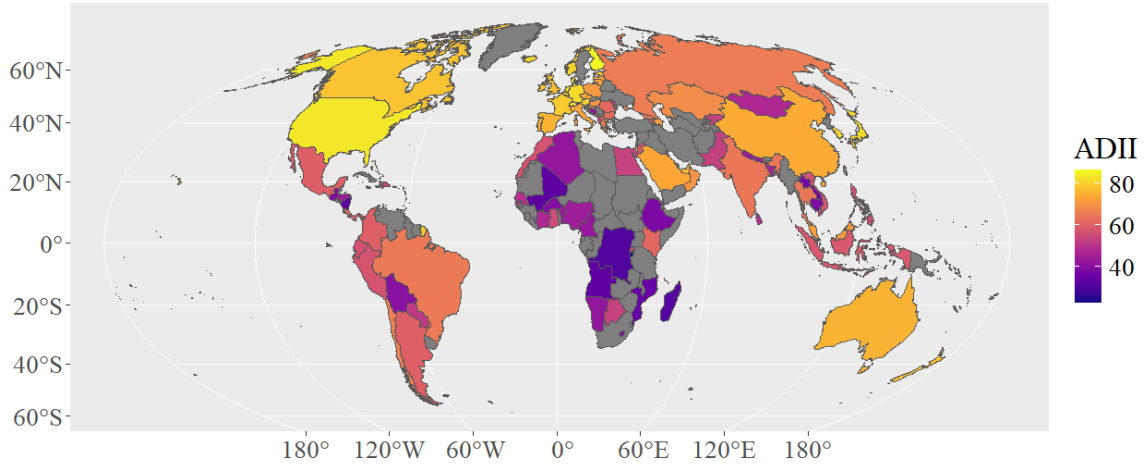


Figure 1: Reconstructed ADII scores. Countries with unavailable data are in grey,

Data availability

Full dataset is available in the Zenodo repository at <https://doi.org/10.5281/zenodo.21216158>

The code in R used to reconstruct the scores and perform the analysis can be found in the GitHub repository at <https://github.com/agachocz/ADII.git>

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